

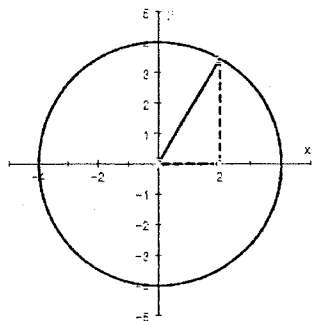
UNIT #5

Exploring Derivatives of Sinusoidal and Exponential Functions

The BIG Trig Sheet

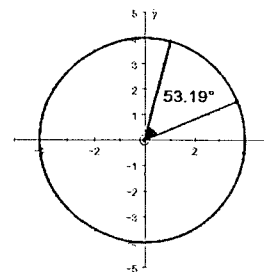
Circle Geometry

1. $\sin(\theta) = \frac{y}{r}$
2. $\cos(\theta) = \frac{x}{r}$
3. $\tan(\theta) = \frac{y}{x}$



Arc Length//Sector Area

1. $s = r\theta$
2. $A = \frac{1}{2}r^2\theta$



Special Angle Chart

Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
Degrees	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$ or $\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$ or $\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$+\infty$

Co-Function Identities for Quadrant I and II

1. $\sin\left(\frac{\pi}{2} - \theta\right) = \cos(\theta)$
2. $\cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$
3. $\tan\left(\frac{\pi}{2} - \theta\right) = \cot(\theta)$
4. $\csc\left(\frac{\pi}{2} - \theta\right) = \sec(\theta)$
5. $\sec\left(\frac{\pi}{2} - \theta\right) = \csc(\theta)$
6. $\cot\left(\frac{\pi}{2} - \theta\right) = \tan(\theta)$
7. $\sin\left(\frac{\pi}{2} + \theta\right) = \cos(\theta)$
8. $\cos\left(\frac{\pi}{2} + \theta\right) = -\sin(\theta)$
9. $\tan\left(\frac{\pi}{2} + \theta\right) = -\cot(\theta)$
10. $\csc\left(\frac{\pi}{2} + \theta\right) = \sec(\theta)$
11. $\sec\left(\frac{\pi}{2} + \theta\right) = -\csc(\theta)$
12. $\cot\left(\frac{\pi}{2} + \theta\right) = -\tan(\theta)$

Compound Angle Formulas

1. $\sin(x + y) = \sin(x)\cos(y) + \sin(y)\cos(x)$
2. $\sin(x - y) = \sin(x)\cos(y) - \sin(y)\cos(x)$
3. $\cos(x + y) = \cos(x)\cos(y) - \sin(x)\sin(y)$
4. $\cos(x - y) = \cos(x)\cos(y) + \sin(x)\sin(y)$

Symmetry Formulas

1. $\sin(-x) = -\sin(x)$
2. $\cos(-x) = \cos(x)$
3. $\tan(-x) = -\tan(x)$

Double Angle Formulas

1. $\sin(2x) = 2\sin(x)\cos(x)$
2. $\cos(2x) = \cos^2(x) - \sin^2(x)$
4. $\cos(2x) = 1 - 2\sin^2(x)$
5. $\cos(2x) = 2\cos^2(x) - 1$

Pythagorean Identities

1. $\sin^2(\theta) + \cos^2(\theta) = 1$
2. $1 + \cot^2(\theta) = \csc^2(\theta)$
3. $\tan^2(\theta) + 1 = \sec^2(\theta)$

Quick Trig Review Questions

Steps to solve a linear trigonometric equation algebraically:

1. Determine the **Related Acute Angle** (Reference Angle β) by using the absolute value of the trig ratio
2. Use the sign of the ratio to determine the quadrant(s) the angle will be in
 - Make a **CAST** diagram
3. Determine the unknown angle(s) within the specified domain.

Ex) Solve for θ if $0 \leq \theta \leq 2\pi$

a) $\cos(\theta) = -\frac{\sqrt{3}}{2}$

b) $\sin(\theta) = \frac{1}{\sqrt{2}}$

c) $\tan(\theta) = -\sqrt{3}$

d) $\cos(\theta) = 0.716$

Steps to find the exact values of a simple trigonometric expression:

1. Get rid of any negative signs by using the symmetry properties of sin, cos, and tan.
2. Determine the **Related Acute Angle** (Reference Angle β) by looking at the denominator. Special angles must have a denominator of 2, 3, 4, or 6.
3. Find out what quadrant your angle is in.
4. Use CAST and knowledge of the reference angle to determine the answer.

Ex) State the exact value for each of the following

a) $\sin\left(\frac{7\pi}{4}\right)$

b) $\cos\left(-\frac{2\pi}{3}\right)$

c) $\tan\left(-\frac{5\pi}{6}\right)$

Determining the Derivative of Cos(x)

To determine the derivative of $y = \cos(x)$ we will use the fact that $\frac{d}{dx} \sin(x) = \cos(x)$

Using our Quadrant 1 Identities, we know that $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$

Differentiate this function using the identity above & the chain rule:

$$\frac{d}{dx} \cos(x) = \frac{d}{dx} \sin\left(\frac{\pi}{2} - x\right)$$

Note: All of the differentiation rules (sum/difference/product/chain) that we have learned before apply to sinusoidal functions ☺

Example Problems

1. Differentiate with respect to θ

a) $y = \sin(\theta) + \cos(\theta)$

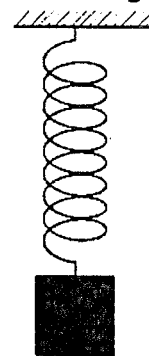
b) $y = 3\pi \sin(\theta) - \frac{\pi}{2} \cos(\theta)$

c) $y = -\pi \cos(\theta) - 8\theta^2 + 2\pi$

2. Determine the slope of the graph of $y = 3 \sin(x)$ at $x = \frac{\pi}{6}$

3. What is the 9th derivative of $y = \cos(x)$?

4. A mass attached to a spring is mounted underneath a table that is 2 metres high. The mass is brought to the top of the table and then released. Its position s at time t can be found using $s(t) = 2 \cos(t)$.

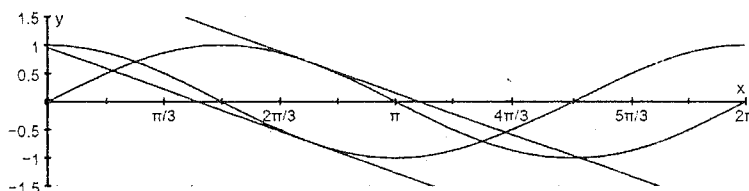


a) For what values of t does the object change its direction?

b) What is its maximum velocity?

5. What is the equation of the normal line to the graph $y = -5\cos(x)$ at $x = \frac{\pi}{4}$

6. Are there any numbers x , $0 \leq x \leq 2\pi$, for which the tangents of $f(x) = \sin(x)$ and $f(x) = \cos(x)$ are parallel? If so, find the values. $\left[\pi - \frac{\pi}{4}, 2\pi - \frac{\pi}{4}\right]$



Warm-Up

1. Determine the equation of the tangent line to the curve $y = 2\pi \sin(x) - \frac{1}{2}$ at $x = \frac{\pi}{6}$.

2. When is the slope of the tangent line to the curve $f(\theta) = \frac{1}{2}\cos(\theta)$ equal to $\frac{\sqrt{3}}{4}$ where $0 \leq \theta \leq 2\pi$.

Derivatives of Sinusoidal functions involving the Product and Chain Rule

Differentiate each of the following

a) $y = \sin(3x)$

b) $y = \sin(x^3)$

c) $y = \sin^3(x)$

d) $y = \sin^3(x^2 - 1)$

e) $y = 2x^3 \cos(x)$

f) $y = 3x^2 \sin(x^3)$

g) $y = 4\sqrt{\cos(2x)}$

h) $y = \sin(\cos(x^2))$

i) $y = \sin(x^2) \cos(3x)$

j) $y = \frac{\sin(x)}{\cos(x)}$

Ex) Find the slope of the function $y = \frac{\cos(2x)}{\sin(x)}$ at $x = \frac{\pi}{2}$

Ex) Find the equation of the tangent line to the curve $f(x) = \sin^2(x)$ at $x = \frac{\pi}{6}$

Ex) For what values of x does the graph of $f(x) = x + 2 \sin(x)$ have a horizontal tangent?

Trig Derivatives Worksheet

Find $\frac{df}{dx}$ for the following functions:

1. $f(x) = 3\cos(4x)$

2. $f(x) = \cos(2x^3)$

3. $f(x) = \cos^3(2x)$

4. $f(x) = (x + \cos(x))^2$

5. $f(x) = (\sin(2x) + \cos(x))^3$

6. $f(x) = x^2 \cos(3x^3)$

7. $f(x) = \sin(x) \sin(2x) \sin(3x)$

8. $f(x) = \frac{\sin(2x)}{\cos(2x)}$

9. $f(x) = \frac{\sin(x^2)}{\cos(2x)}$

10. $f(x) = \frac{\sin(2x)}{1+\cos(2x)}$

11. $f(x) = \frac{1+\sin(x)}{x+\cos(x)}$

12. Find the 27th derivative of $f(x) = \cos(x)$

13. $f(x) = \sqrt{x^2 + \sin(3x)}$

14. $f(x) = \frac{\sin(x)}{\cos(x)+2}$

15. $f(x) = \sin(\cos(x^2))$

Answers:

1. $f'(x) = -12 \sin(4x)$

2. $f'(x) = -6x^2 \sin(2x^3)$

3. $f'(x) = -6 \cos^2(2x) \sin(2x)$

4. $f'(x) = 2x + 2 \cos(x) - 2x \sin(x) - 2 \cos(x) \sin(x) = 2(x + \cos(x))(1 - \sin(x))$

5. $f'(x) = 3(\cos(x) + \sin(2x))^2(2 \cos(2x) - \sin(x))$

6. $f'(x) = 2x \cos(3x^3) - 9x^4 \sin(3x^3)$

7. $f'(x) = \cos(x) \sin(2x) \sin(3x) + 2 \cos(2x) \sin(x) \sin(3x) + 3 \cos(3x) \sin(x) \sin(2x)$

8. $f'(x) = 2 \sec^2(2x)$

9. $f'(x) = 2x \cos(x^2) \sec(2x) + 2 \tan(2x) \sec(2x) \sin(x^2)$

10. $f'(x) = \frac{2}{1+\cos(2x)} = \sec^2(x)$

11. $f'(x) = \frac{x \cos(x)}{(x+\cos(x))^2}$

12. $f^{(27)}(x) = \sin(x)$

13. $f'(x) = \frac{1}{2}(x^2 + \sin(3x))^{-\frac{1}{2}}(2x + 3 \cos(3x))$

14. $f'(x) = \frac{1+2 \cos(x)}{(\cos(x)+2)^2}$

15. $f'(x) = -2x \cos[\cos(x^2)] \sin(x^2)$

Examples: Determine the derivative of the following:

a) $y = xe^x$

b) $y = e^{x^2} \sin x^2$

c) $f(x) = \frac{x^2}{e^{2x+1}}$

d) $y = \sin(e^{x^2}) \cos(e^{x^2})$

e) $y = e^{\sqrt{5x^2+3}}$

f) $y = \sqrt{x + e^{1-x^2}}$

Ex) Differentiate the following

a) $y = x \ln(x)$

b) $y = \ln(3x^2 - 5x + 1)$

The Chain Rule for the Natural Logarithm: $\frac{d}{dx} \ln[g(x)] = \frac{1}{g(x)} \cdot g'(x)$

Ex) Find the derivatives of the following functions

a) $y = \ln(x^2 + 3)$

b) $y = \ln(\cos(x))$

c) $y = (\ln(x) + 2x)^3$

d) $y = \ln\left(\frac{x}{\sqrt{x+1}}\right)$ Hint: Use the Laws of logarithms! $\log\left(\frac{a}{b}\right) = \underline{\hspace{2cm}}$

e) $y = \ln\left(\sqrt{\frac{x}{3x-5}}\right)$

1. Differentiate the following

a) $f(x) = \ln(9 - x^2)$

b) $f(x) = \frac{\ln(x^2)}{x^3}$

c) $f(x) = \frac{\ln(\sin(x))}{\cos(x)}$

d) $f(x) = \ln\left(\frac{(x^2+3)(3x-5)^7}{5x-3}\right)$ (do not simplify)

Ex) Find the derivative of $y = 5 \left(\frac{1}{4}\right)^x$.

Note: Will never get negative b/c $\ln(-1) = i\pi$

Ex) Find the derivative of $y = 3^{x^2+5}$

Ex) Find the equation of the tangent line to the curve $y = 8^x$ at $x = 3$

Ex) A bacterial culture's population is tripling every 5 hours. There are 9001 bacteria when the culture is discovered.

a) Write an equation that relates the number of bacteria to the time $N = N_0(R)^{\frac{t}{a}}$

b) Determine the number of bacteria after 8 hours

c) How fast is the bacteria population increasing after 8 hours?

Ex) Differentiate $y = (2x^2 + 5)^7$ using the chain rule.

Ex) Differentiate $y = (2x^2 + 5)^7$ using logarithmic differentiation.

Ex) Find the derivative of $y = x^{\sin(x)} \left[\frac{dy}{dx} = x^{\sin(x)} \left(\cos(x) \ln(x) + \frac{\sin(x)}{x} \right) \right]$

Practice Derivative Questions

1. $y = 3^{x^2-7} \cos(3x)$

2. $y = \log_7(\sin(x))$

3. $y = 3^{2x} \tan(x)$

4. $y = \frac{12x^2}{\log_4(3x^6)}$

5. $y = \ln(1 + e^{5x})$

6. $y = \frac{\sin(2x)}{1+\sin(2x)}$

7. $y = x^3 \log_4(x^5 - 3x + 1)$

8. $y = 4^{x^3-5x} \log_4(2x + 5)$

9. $y = \sin(x^2) \log_2(x^2)$

10. Determine the x coordinate where the slope of the tangent line to $y = \log_2(5x + 1)$ is equal to zero

Answers

1. $y' = 3^{x^2-7} [2x \ln(3) \cos(3x) - 3 \sin(3x)]$

2. $y' = \frac{\cot(x)}{\ln(7)}$

3. $y' = 3^{2x} (\sec^2(x) + \ln(9) \tan(x))$

4. $y' = \frac{24x [\ln 4 \log_4(3x^6) - 3]}{\ln 4 [\log_4(3x^6)]^2}$

5. $y' = \frac{5e^{5x}}{e^{5x} + 1}$

6. $y' = \frac{2 \cos(2x)}{(\sin(2x) + 1)^2}$

7. $y' = x^2 \left[3 \log_4(x^5 - 3x + 1) + \frac{x(5x^4 - 3)}{(x^5 - 3x + 1) \ln 4} \right]$

8. $y' = 4^{x^3-5x} \left[(\ln 4)(3x^2 - 5) \log_4(2x + 5) + \frac{2}{(2x + 5) \ln(4)} \right]$

9. $y' = 2x \left[\cos x^2 \cdot \log_2 x^2 + \frac{\sin x^2}{x^2 \ln 2} \right]$

10. $\left(\frac{5 - \ln(2)}{5 \ln(2)}, ? \right)$

FRONT DERIVATIVES OF EXPONENTIAL FUNCTIONS

4. Differentiate.

- (a) $y = 2e^{-x}$ (b) $y = x^4 e^x$
 (c) $y = e^{2x} \sin 3x$ (d) $y = e^{\sqrt{x}}$
 (e) $y = e^{\tan x}$ (f) $y = \tan(e^x)$
 (g) $y = \frac{e^x}{x}$ (h) $y = \frac{e^x}{1 - e^{2x}}$
 (i) $y = e^{\sin(x^2)}$ (j) $y = x e^{\cot 4x}$
 (k) $y = (1 + 5e^{-10x})^4$ (l) $y = \sqrt{x + e^{-1-x^2}}$

5. Find the equation of the tangent line to the curve $y = 1 + x e^{2x}$ at the point where $x = 0$.

6. Find y' if $e^{xy} = 2x + y$.

7. If $f(x) = e^{2x}$, find $f^{(6)}(0)$.

8. Find the intervals of increase and decrease for the function $f(x) = x^2 e^{-x}$.

9. Find the absolute minimum value of the function $f(x) = \frac{e^x}{x}$, $x > 0$.

10. For the function $f(x) = x e^x$, find

- (a) the absolute minimum value,
 (b) the intervals of concavity,
 (c) the inflection point.

10. (a) $-\frac{e}{1}$ (b) $\frac{e}{2}$ on $(-\infty, -2)$, $\frac{e}{2}$ on $(-2, \infty)$, $\frac{e}{2}$ on $(-\infty, -2)$
 9. e
 8. increasing on $(0, 2)$, decreasing on $(-\infty, 0)$, $(2, \infty)$
 5. $x - y + 1 = 0$
 6. $\frac{2}{x^2} - \frac{1}{y e^{xy}}$
 7. 64
 (l) $y' = \frac{2\sqrt{x+e^{-1-x^2}}}{1-2xe^{2x}}$
 (k) $y' = -200e^{-10x}(1+5e^{-10x})^3$
 (j) $y' = e^{\cot 4x}(1-4x \csc^2 4x)$
 (i) $y' = 2x \cos(x^2) e^{\sin(x^2)}$
 (h) $y' = \frac{1}{1+e^{2x}}$
 (f) $y' = \frac{e^x}{e^x + 1}$
 (g) $y' = \frac{e^x}{x^2} = \frac{1}{x^2} e^x$
 (d) $y' = \frac{2\sqrt{x}}{e^x}$
 (e) $y' = \frac{e^x}{x^2} (\sec^2 x) e^{\tan x}$
 (c) $y' = e^{2x}(2 \sin 3x + 3 \cos 3x)$
 (a) $y' = -2e^{-x}$ (b) $y' = x^2 e^{(4+x)}$

Answers

WORKING WITH LOGARITHMS

3. Evaluate without using a calculator.

- (a) $e^{\ln 5}$ (b) $\ln e^2$
 (c) $2 \ln e$ (d) $e^{\ln 2}$
 (e) $\ln \sqrt{e}$ (f) $\ln 2 + 2 \ln 3 - \ln 18$

4. Solve for x .

- (a) $e^x = 4$ (b) $\ln x = 6$
 (c) $\ln(2x - 1) = 1$ (d) $e^{3x+5} = 10$
 (e) $\ln(e^{3-x}) = 8$ (f) $\ln x = \ln 4 + \ln 7$
 (g) $\ln(\ln x) = 2$ (h) $e^{e^x} = 5$

5. Find the solution of each equation correct to six decimal places.

- (a) $\ln(x + 1) = 3$ (b) $e^{-x} = \frac{1}{2}$
 (c) $e^{5x+3} = 10$ (d) $2^{x-5} = 3$

6. Express as a single logarithm.

- (a) $\frac{1}{3} \ln x + 2 \ln(3x - 5)$
 (b) $2 \ln x - \frac{1}{2} \ln(x^2 - 1) + 3 \ln(x^2 + 1)$

6. (a) $\ln \ln \left(\frac{\sqrt{x^2+1}}{x^2(x^2+1)^2} \right)$ (b) $[-5, 2]$
 (c) -0.139483 (d) 6.584963
 5. (a) 19.085537 (b) 0.693147
 (g) e^{e^2} (h) $\ln(\ln 5)$
 (d) $\frac{1}{3} \ln 10 - 5$ (e) -5 (f) 28
 4. (a) $\ln 4$ (b) e^e (c) $\frac{1}{2}(\ln 4 + 1)$
 3. (a) 5 (b) 2 (c) 2 (d) 32 (e) $\frac{7}{2}$ (f) 0

Answers

BACK: DERIVATIVES OF LOGARITHMIC FUNCTIONS

1. Differentiate.

- (a) $f(x) = x^2 \ln x$ (b) $f(x) = \sqrt{\ln x}$
 (c) $g(x) = \ln(x^3 + 1)$ (d) $g(x) = \ln(5x)$
 (e) $y = \sin(\ln x)$ (f) $y = \ln(\sin x)$
 (g) $y = \frac{\ln x}{x^3}$ (h) $y = (x + \ln x)^3$

- (i) $y = \ln|2x + 1|$ (j) $y = \ln\left(\frac{x+1}{x-1}\right)$
 (k) $y = \ln \sqrt{\frac{x}{2x+3}}$ (l) $y = \ln \frac{x}{\sqrt{x^2+1}}$
 (m) $y = \ln(\sec x + \tan x)$ (n) $y = \tan[\ln(1 - 3x)]$

2. (a) If $f(x) = \ln(\ln x)$, find $f'(x)$.
 (b) Find the domains of f and f' .

3. Find the derivative of each function.

- (a) $f(x) = \log_2(x^2 + 1)$ (b) $g(x) = x \log_{10} x$
 (c) $F(x) = \log_5(3x - 8)$ (d) $G(x) = \frac{1 + \log_3 x}{x}$

4. Differentiate.

- (a) $y = x^3 + 3^x$ (b) $y = 2^{x^2-1}$
 (c) $y = x5^{\sqrt{x}}$ (d) $y = 10^{\ln \pi x}$

5. Find the equation of the tangent line to each curve at the given point.

- (a) $y = \ln(x - 1)$, $(2, 0)$ (b) $y = x^2 \ln x$, $(1, 0)$
 (c) $y = 10^x$, $(1, 10)$ (d) $y = \log_{10} x$, $(100, 2)$

6. Find y' if $\ln(x + y) = y - 1$.

7. (a) Find the absolute minimum value of the function $f(x) = x \ln x$.
 (b) On what interval is it concave upward?

8. (a) Find the local maximum and minimum values of the function $f(x) = x(\ln x)^2$.
 (b) Find the inflection point of this function.

1. (a) $f'(x) = x(2 \ln x + 1)$ (b) $f'(x) = \frac{2x\sqrt{\ln x}}{1 + x^2}$ (c) $g'(x) = \frac{1}{x} \cos(\ln x)$ (d) $g'(x) = \frac{x}{1} \cos(\ln x)$ (e) $y' = \frac{x}{1} \cos(\ln x)$ (f) $y' = \frac{x}{1} \cos(\ln x)$ (g) $y' = \frac{x}{1} \cos(\ln x)$ (h) $y' = \frac{x}{1} \cos(\ln x)$ (i) $y' = \frac{2x+1}{2}$ (j) $y' = -\frac{x^2-1}{2}$ (k) $y' = \frac{2x(2x+3)}{3}$ (l) $y' = \frac{x(x^2+1)}{1}$ (m) $y' = \sec x$ (n) $y' = \frac{1-3x}{-3\sec^2[\ln(1-3x)]}$
 2. (a) $f'(x) = \frac{1}{x \ln x}$ (b) $(1, \infty)$, $(1, \infty)$
 3. (a) $f'(x) = \frac{(x^2+1) \ln 2}{2x}$ (b) $g'(x) = \log_{10} x + \frac{1}{x}$ (c) $F'(x) = \frac{(3x-8) \ln 5}{3}$ (d) $G'(x) = \frac{(\ln 3)x^2}{1 - \ln x - \ln 3}$ (e) $y' = 3x^2 + 3^x \ln 3$ (f) $y' = (4x^2 - 1)(\ln 2)2^{x^2-1}$ (g) $y' = \frac{2}{5\sqrt{x}}(\sqrt{x} \ln 5 + 2)$ (h) $y' = \pi(\sec^2 \pi x) \ln 10(10^{\pi x})$
 4. (a) $x - y - 2 = 0$ (b) $x - y - 1 = 0$ (c) $10(\ln 10)x - y - 10(\ln 10 - 1) = 0$ (d) $x - y - 100(\ln 10)y + 100(2 \ln 10 - 1) = 0$
 5. (a) $\frac{1}{x+y-1}$ (b) $(0, \infty)$
 6. (a) local maximum $f(e^{-2}) = 4e^{-2}$, local minimum $f(1) = 0$ (b) $(\frac{e}{1}, \frac{e}{1})$

Answers

2. Find the equation of the tangent line to the curve $y = e^x + \ln(x)$ at $x = 2$. (exact numbers only)

3. What is the equation of the tangent line to the curve $y = x + \cos(x)$ at $x = \frac{7\pi}{6}$

4. What is the equation of the normal line to the curve $y = \log_3(x^3 - 2x)$ at the point $x = 2$

5. Determine the values of x , $0 \leq x \leq 2\pi$ where the graph of $y = \frac{\sqrt{3}}{2}\cos(x) - \frac{1}{2}\cos^2(x)$ has a horizontal tangent line.

6. Determine any local maxima or local minima for $y = x^2 e^x$ $\left[\text{max: } \left(-2, \frac{4}{e^2}\right) \text{ min: } (0, 0) \right]$

7. Determine the derivative of $y = 5^{\cos(x)} \sin^2(x^2 - 3)$

Practice: Sinusoidal, Exponential and Logarithmic Functions

1. Differentiate and simplify fully.

a) $f(t) = \sqrt{2 + \sin^2 5t}$

b) $h(x) = \frac{x^2}{2 - \cos \pi x}$

c) $f(t) = 3e^{\sin 2t}$

d) $g(x) = e^{\sqrt{x}} \ln \sqrt{x}$

e) $y = 2^x \log_2(x^4)$

f) $h(u) = \frac{e^{3u}}{1 + e^{3u}}$

2. Find the maximum and minimum value of $f(x) = 2 \sin x - x$, $0 \leq x \leq 2\pi$.

3. Find the maximum and minimum value of $f(x) = xe^{-2x}$ on the interval $0 \leq x \leq \ln 3$.

4. Determine the point(s) of inflection for $f(w) = \frac{\ln w^2}{w}$.

Derivatives of Exponential and Logarithmic Functions Review

1. Differentiate each of the following

a) $y = e^{x \cos(x)}$

$$[y' = e^{x \cos(x)} (\cos(x) - x \sin(x))]$$

b) $y = \sin^2(\ln(x))$ *double angle formula

$$[y' = \frac{\sin(2 \ln(x))}{x}]$$

c) $y = \ln\left(\frac{(2x+5)^3}{(2x-7)^2(x^2-4)}\right)$ (don't simplify)

$$[y' = \frac{-2(2x^3+27x^2-27x-124)}{(2x-7)(2x+5)(x^2-4)}]$$

d) $y = \log_5(x^3 e^x)$

$$[y' = \frac{x+3}{x \ln(5)}]$$

e) $y = x^5 \log(x^3 + 7x - 5)$

$$[y' = \frac{x^4}{\ln 10} (5 \ln(x^3 + 7x - 5) + \frac{x(3x^2+7)}{x^3+7x-5})]$$

f) $y = \cos(2x) \sin^2(x)$

$$[y' = \sin(2x) (2 \cos(2x) - 1)]$$

g) $y = 7^{x^3+2x} (5x^3 - 2x)$

$$[y' = 7^{x^3+2x} (15x^5 \ln(7) + 4x^3 \ln(7) + 15x^2 - 4x \ln(7) - 2)]$$

2. Evaluate

a) $\sin\left(\frac{4\pi}{3}\right)$

b) $\cos\left(\frac{5\pi}{6}\right)$

3. Solve where $0 \leq \theta \leq 2\pi$

a) $\sin(\theta) = -\frac{\sqrt{3}}{2}$

b) $\tan(\theta) = \frac{1}{\sqrt{3}}$

4. What is the equation of the tangent line to the curve $y = x^3 \ln(x)$ at $x = e$? $[y = 4e^2x - 3e^3]$

5. Find an equation of the tangent line to the curve $y = \sec(x) - 2\cos(x)$ at $x = \frac{\pi}{3}$.

$$[y = 3\sqrt{3}x + (1 - \sqrt{3}\pi)]$$

6. Find the x-coordinates of the curve $y = \ln(2x^2 - 3x)$ where the tangent line has a slope of $-\frac{5}{9}$. $[x = -3, \frac{9}{10}]$

7. Determine the coordinates where $y = x^2 \log_3(x)$ has a horizontal tangent line. $[\frac{1}{\sqrt{e}}, -\frac{1}{e \ln(9)}]$

8. Determine the local maxima and minima for the function $y = \frac{x^2}{\ln(x)}, x \geq 0$. $[min: (\sqrt{e}, 2e)]$

Challenge:

9. Determine the derivative of $y = \tan^{-1}(x)$. Hint: Differentiate $\tan(y) = x$ using implicit differentiation. You will have to use a trigonometric identity afterwards. $[\frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2}]$